Francisco Pereira

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Education

- **Instituto Superior Técnico (University of Lisbon)** — Lisbon, Portugal
  PhD student - Computer Science and Engineering
  Anticipated graduation date: 09/26 — 2021 - Present
- **Instituto Superior Técnico (University of Lisbon)** — Lisbon, Portugal
  MSc - Computer Science and Engineering — 2019 - 2021
  - **Specializations**: Distributed Systems and Cyber-Security
  - **Thesis**: Parallel NF Synthesis – NFV field, supervised by Professor Luís Pedrosa
- **Instituto Superior Técnico (University of Lisbon)** — Lisbon, Portugal
  BSc - Computer Science and Engineering — 2016 - 2019

Projects

- **SyNAPSE - Synthesizing Network Accelerators using Programmable Switching Equipment**:
  - **Description**: When implementing network functions (NFs), developers are often confronted with a choice: implement the NF in software and face the challenge of performance, or use one of a wide variety of programmable networking devices, such as programmable switches and SmartNICs, to trade-off some flexibility for the ability to process packets at full line rate. With SyNAPSE we ask "Why not have both?". We propose a synthesis based approach to automatically generate accelerated implementations of a Software NFs, using smart network devices whenever possible to increase performance.
  - **Context**: INESC-ID, FCiências.ID (Associação para a Investigação e Desenvolvimento de Ciências), and the Department of Computer Science of Carnegie Mellon University
  - **PIs**: Luís Pedrosa (UL), Fernando Ramos (UL), Justine Sherry (CMU), Ruben Martins (CMU).
- **Maestro - Automatic Parallelization of Software Network Functions**:
  - **Description**: Software network functions (NFs) trade-off flexibility and ease of deployment for an increased challenge of performance. Fortunately, a number of tools, techniques, and architectural enhancements have come together over the years to enable packet processing at 10s Gbps, but achieving these traffic rates requires a deep understanding of low-level details. The traditional way to increase NF performance is by distributing traffic to multiple CPU cores, but this poses a significant challenge: *how to parallelize an NF without breaking its semantics?* We propose Maestro, a tool that analyzes a sequential implementation of an NF and automatically generates an enhanced parallel version.
    Maestro is currently under submission, and available upon request.

Publications

- **Automatic Parallelization of Software Network Functions** — 2024
  Francisco Pereira, Fernando Ramos, Luis Pedrosa.
  21st USENIX Symposium on Networked Systems Design and Implementation (**NSDI'24**)
- **Automatic Generation of Network Function Accelerators Using Component-Based Synthesis** — 2022
  Francisco Pereira, Gonçalo Matos, Hugo Sadok, Daehyeok Kim, Ruben Martins, Justine Sherry, Fernando Ramos, Luis Pedrosa.
  ACM SIGCOMM Symposium on SDN Research (**SOSR'22**)

Scholarships and Awards

- CMU Portugal Affiliate Ph.D. Program, FCT (Fundação para as Ciências e Tecnologias) — 2021-2025
- Master Student Merit Award 2020, Distributed Systems Group @ INESC-ID — 2020

Extracurricular Activity

- **Research Assistant Intern @ Carnegie Mellon University** — Pittsburgh, Pennsylvania, USA
  CS Research Department – School of Computer Science — 2022
- **Netdev 0x16** — Lisbon, Portugal
  Organization committee — 2022
- **EuroSys'22** — Virtual
  Shadow PC — 2021
- **The Networking Channel** — Virtual
  Student panelist – Networking Education During and After the Pandemic — 2021
- **EuroP4'21** — Virtual
  HotCRP chair — 2021
- **SIGCOMM'21** — Virtual
  Student volunteer — 2021
- **SINFO - Semana da Informática do Instituto Superior Técnico** — Lisbon, Portugal
  Team Leader @ DevTeam — 2018-2020

Teaching

- **Distributed Systems** — IST, Portugal
  Teaching Assistant — Spring 2022
- **Computer Networks** — IST, Portugal
  Teaching Assistant — Fall 2022
- **Computer Networks** — IST, Portugal
  Teaching Assistant — Fall 2021

Programming skills

- **Programming Languages**:
  - **Very comfortable**: C, C++, Python (2 & 3), Javascript
  - **Comfortable**: Java
  - **Used in the past**: C#, Go
- **Frameworks**: DPDK, KLEE, LLVM, Z3, node.js, AngularJS, Gorilla (Go)
- **Other tools**: Docker, vagrant, gdb, git, bash, gnuplot